Computerized Mathematics Assignment

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Question 1

3 marks

Given: p: “Eric reads Newsweek” · q: “Eric reads the New Yorker” · r: “Eric reads Times”

a) Eric reads Newsweek and the New Yorker but not Times

$$p \wedge q \wedge \neg r$$

b) Eric reads Newsweek and the New Yorker, or he does not read Newsweek and Time

$$(p \wedge q) \vee \neg(p \wedge r)$$

c) It is not true that Eric reads Time and the New Yorker but not Newsweek

$$\neg((r \wedge q) \wedge \neg p)$$

Question 2

8 marks

a) Truth table for $p \wedge \neg q$

p	q	$\neg q$	$p \wedge \neg q$
T	T	F	**F**
T	F	T	**T**
F	T	F	**F**
F	F	T	**F**

b) Truth table for $\neg p \vee \neg q$

p	q	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	F	F	**F**
T	F	F	T	**T**
F	T	T	F	**T**
F	F	T	T	**T**

c) Truth table for $\neg(\neg p \wedge q)$

p	q	$\neg p$	$\neg p \wedge q$	$\neg(\neg p \wedge q)$
T	T	F	F	**T**
T	F	F	F	**T**
F	T	T	T	**F**
F	F	T	F	**T**

d) Truth table for $\neg(\neg p \wedge \neg q)$

p	q	$\neg p$	$\neg q$	$\neg p \wedge \neg q$	$\neg(\neg p \wedge \neg q)$
T	T	F	F	F	**T**
T	F	F	T	F	**T**
F	T	T	F	F	**T**
F	F	T	T	T	**F**

Question 3

5 marks

Proposition: $\sim(p \wedge q) \wedge (p \vee q)$

p	q	$p \wedge q$	$\sim(p \wedge q)$	$p \vee q$	$\sim(p \wedge q) \wedge (p \vee q)$
T	T	T	F	T	**F**
T	F	F	T	T	**T**
F	T	F	T	T	**T**
F	F	F	T	F	**F**

Conclusion: For the proposition as written, the final column yields $(\mathrm{F}, \mathrm{T}, \mathrm{T}, \mathrm{F})$. Therefore, it is a **contingency** (equivalent to exclusive-OR, $p \oplus q$).

Question 4

4 marks

Proof of the printed identity: $p \vee (q \vee r) \equiv (p \vee q) \vee r$

p	q	r	$q \vee r$	**p** $\vee$ (**q** $\vee$ **r**)	$p \vee q$	(**p** $\vee$ **q**) $\vee$ **r**
T	T	T	T	**T**	T	**T**
T	T	F	T	**T**	T	**T**
T	F	T	T	**T**	T	**T**
T	F	F	F	**T**	T	**T**
F	T	T	T	**T**	T	**T**
F	T	F	T	**T**	T	**T**
F	F	T	T	**T**	F	**T**
F	F	F	F	**F**	F	**F**

Conclusion: Since the columns $p \vee (q \vee r)$ and $(p \vee q) \vee r$ have identical truth values for all combinations, the equivalence is proven.

Note: For the distributive law described in words, $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$, both sides evaluate identically to $\mathrm{T}, \mathrm{T}, \mathrm{T}, \mathrm{T}, \mathrm{T}, \mathrm{F}, \mathrm{F}, \mathrm{F}$.

Question 5

4 marks

Normalized form

Number	Normalized form (fractional: $0.1 \le \lvert m\rvert < 1$)	Normalized form (scientific: $1 \le \lvert m\rvert < \text{base}$)
a) 1001.1010_2	$0.10011010_2 \times \mathbf{2^4}$	$1.0011010_2 \times 2^3$
b) 0.007633	$0.7633 \times \mathbf{10^{-2}}$	7.633×10^{-3}
c) 444.222	$0.444222 \times \mathbf{10^3}$	4.44222×10^2
d) 11011.1100_2	$0.110111100_2 \times \mathbf{2^5}$	$1.10111100_2 \times 2^4$

Question 6

4 marks

a) $||2-5|-|4-11||$

$$= ||-3|-|-7|| = |3-7| = |-4| = \mathbf{4}$$

b) $|2-7|-|4-1|$

$$= |-5|-|3| = 5-3 = \mathbf{2}$$

Question 7

6 marks

Given true value $A = 55.777$.

a) A is rounded to 55.78

- Absolute error:

$$e = |55.777 - 55.78| = |-0.003| = 0.003$$

- Relative error:

$$e_r = \frac{e}{|A|} = \frac{0.003}{55.777} \approx 0.000053786 \quad (5.38 \times 10^{-5})$$

b) A is truncated to 55.77

- Absolute error:

$$e = |55.777 - 55.77| = 0.007$$

- Relative error:

$$e_r = \frac{e}{|A|} = \frac{0.007}{55.777} \approx 0.00012550 \quad (1.26 \times 10^{-4})$$

Question 8

8 marks

Given $\mathbb{N} = \{1, 2, 3, \ldots\}$.

a) $A = \{x : x \in \mathbb{N},\ 2 < x < 15\}$

$$A = \{3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$$

b) $B = \{x : x \in \mathbb{N},\ x \text{ is odd},\ x < 15\}$

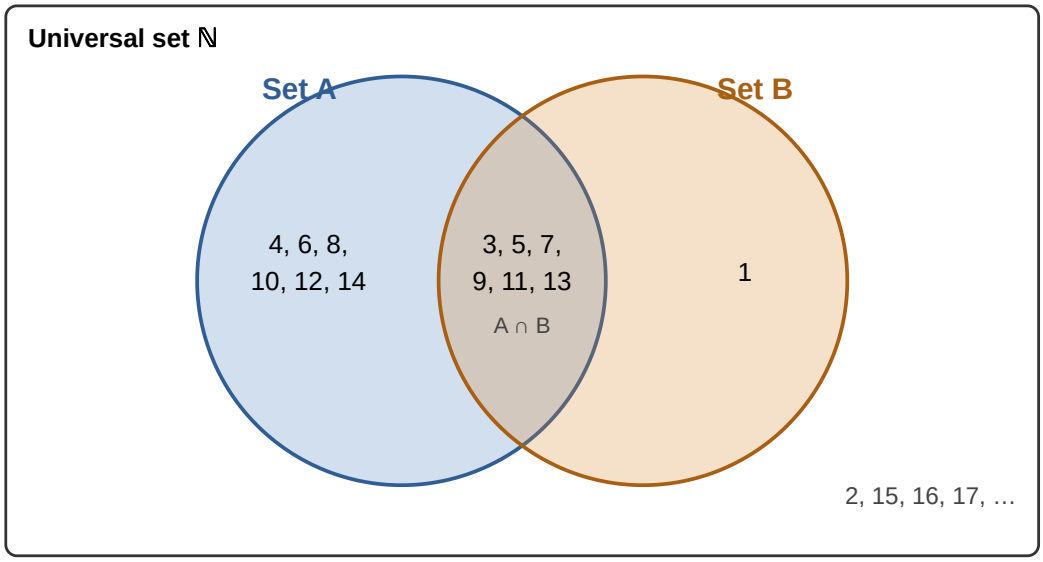
$$B = \{1, 3, 5, 7, 9, 11, 13\}$$

c) $C = \{x : x \in \mathbb{N},\ 4 - x = 2\}$

$$4 - x = 2 \implies x = 2, \qquad C = \{2\}$$

d) Venn diagram for $A \cap B$

- $A \cap B = \{3, 5, 7, 9, 11, 13\}$
- Only in A ($A \setminus B$): $\{4, 6, 8, 10, 12, 14\}$
- Only in B ($B \setminus A$): $\{1\}$



Question 9

12 marks

a) $(0.7125 \times 10^2) \times (0.8421 \times 10^3)$

$$= (0.7125 \times 0.8421) \times 10^{2+3} = 0.59999625 \times \mathbf{10^5} \quad (\text{or } 5.9999625 \times 10^4)$$

b) $(0.5451 \times 10^5) \times (7.342 \times 10^2)$

$$= (0.5451 \times 7.342) \times 10^{5+2} = 4.0021242 \times 10^7 = 0.40021242 \times \mathbf{10^8}$$

c) $0.1245 \times 10^3 + 0.2214 \times 10^5$

Align exponents to 10^5: $0.1245 \times 10^3 = 0.001245 \times 10^5$. Then add the mantissas:

$$= (0.001245 + 0.2214) \times 10^5 = 0.222645 \times \mathbf{10^5} \quad (\text{or } 2.22645 \times 10^4)$$

d) $0.1101_2 \times 2^5 - 0.1010_2 \times 2^5$

Subtract the mantissas: $0.1101_2 - 0.1010_2 = 0.0011_2$. Then normalize:

$$0.0011_2 \times 2^5 = 0.11_{\mathbf{2}} \times \mathbf{2^3} \quad (\text{or } 1.1_2 \times 2^2)$$

Question 10

6 marks

1. $E_1 = xyz + xy'z' + xy'z + x'y'z'$

Minterms: $xyz = 111 = m_7$, $xy'z' = 100 = m_4$, $xy'z = 101 = m_5$, $x'y'z' = 000 = m_0$.

Karnaugh map

$x \setminus yz$	**00**	**01**	**11**	**10**
0	**1** (m_0)	0	0	0
1	**1** (m_4)	**1** (m_5)	**1** (m_7)	0

Grouping

- Group 1: cells $\{m_0, m_4\} \Longrightarrow \mathbf{y'z'}$
- Group 2: cells $\{m_5, m_7\} \Longrightarrow \mathbf{xz}$

Minimal sum:

$$\mathbf{E_1} = \mathbf{y'z'} + \mathbf{xz}$$

2. $E_2 = xy'z + x'yz' + xy'z' + x'yz + x'y'z' + x'y'z$

Minterms: $xy'z = m_5$, $x'yz' = m_2$, $xy'z' = m_4$, $x'yz = m_3$, $x'y'z' = m_0$, $x'y'z = m_1$.

Karnaugh map

$x \setminus yz$	**00**	**01**	**11**	**10**
0	**1** (m_0)	**1** (m_1)	**1** (m_3)	**1** (m_2)
1	**1** (m_4)	**1** (m_5)	0	0

Grouping

- Group 1: entire top row $\{m_0, m_1, m_3, m_2\} \Longrightarrow \mathbf{x'}$
- Group 2: quad block $\{m_0, m_1, m_4, m_5\} \Longrightarrow \mathbf{y'}$

Minimal sum:

$$\mathbf{E_2} = \mathbf{x'} + \mathbf{y'}$$